

Fractional SPDE with Time-independent Noise: Solvability and Exact Asymptotics*

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Abstract

In this article, we study the

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1 Introduction

In this paper we study the following fractional stochastic equation:

$$\begin{cases} (\partial^b + \frac{\nu}{2}(-\Delta)^{\alpha/2}) u(t, x) = I_t^\gamma \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad u_t(0, \cdot) = 0 & b \in (1, 2] \end{cases} \quad (1.1)$$

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where $a \in (0, 2)$, $b \in (0, 2)$, $\nu > 0$, $r > 0$ and $\theta > 0$ and $W = \{W(\varphi) : \varphi \in \mathcal{D}(\mathbb{R}^d)\}$ is a time-independent Gaussian process, defined on a complete probability space $(\Omega, \mathcal{F}, P)$, with mean zero and covariance

$$\mathbb{E}[W(\varphi)W(\psi)] = \int_{\mathbb{R}^d} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi) =: \langle \varphi, \psi \rangle_{\mathcal{H}},$$

with the spectral measure μ being assumed to be a non-negative and non-negative definite tempered measure on $\mathbb{R}^d$. We choose $\mathcal{F}\varphi(\xi) = \int_{\mathbb{R}^d} \exp(-ix\xi) \varphi(x) dx$ to be the Fourier transform of φ . It is known that the Fourier transform of μ , denoted by γ , is also a non-negative and non-negative definite tempered measure. See pages 51 and 52 of [7].

Above, ∂_t^b denotes the Caputo fractional differential operator:

$$\partial_t^b f(t) := \begin{cases} \frac{1}{\Gamma(m-b)} \int_0^t d\tau \frac{f^m(\tau)}{(t-\tau)^{b+1-m}} & \text{if } m-1 < b < m, \\ \frac{d^m}{dt^m} f(t) & \text{if } b = m, \end{cases}$$

and I_t^r is the Riemann-Liouville fractional integral of order $r > 0$:

$$I_t^r f(t) := \frac{1}{\Gamma(r)} \int_0^t (t-s)^{r-1} f(s) ds, \quad \text{for } t > 0,$$

with the convention that $I_t^0 = \text{Id}$ (the identity operator).

The fundamental solutions to 1.1 consist of a triplet:

$$\left\{ Z_{a,b,\nu,d}(t, x), Z_{a,b,\nu,d}^*(t, x), Y_{a,b,r,\nu,d}(t, x) : t > 0, x \in \mathbb{R}^d \right\}.$$

The actual definitions of each member of the above triplet do not play a crucial role throughout this paper and for this reason we postpone further discussion about them to the appendix at the end of this paper. Since we are interested in the constant one initial condition (and 0 initial velocity when $b > 1$), by Theorem 4.1 [3] the corresponding solution to the homogeneous equation (i.e. the solution when there is no driving source) is equal to the constant one. Hence, by setting $G(t, x) = Y_{a,b,r,\nu,d}(t, x)$, 1.1 can be written as the following Skorohod integral equation:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds. \quad (1.2)$$

Our goal is to develop exact moment asymptotics to the solution $u(t, x)$. We follow the techniques laid out in [1] where in that paper they focus exclusively on the stochastic wave equation. This paper acts as a generalization of that paper to the fraction stochastic equation. In fact, the techniques presented in this paper are quite general and can be applied to any SPDE that has a Green's function that satisfies a series of scaling properties which we present throughout this paper. Note that the fractional equation 1.1 recovers the stochastic wave equation when $a = b = 2$ and $r = 0$ and in addition we recover the stochastic heat equation when $b = 1$, $a = 2$ and $r = 0$. We need to make note about restrictions on the spatial dimension due to the fact that the fundamental solution to the wave equation only exists as a function for $d \leq 2$ and because of that reason we restrict ourselves to $d \leq 2$.

In addition, we need to restrict ourselves to the Gaussian Noise with $\mu(dx) = \varphi(x)dx$ with $\varphi(x) = C_{d,\alpha} |x|^{-(d-\alpha)}$ where $C_{d,\alpha} = \pi^{-d/2} 2^{-\alpha} \frac{\Gamma((d-\alpha)/2)}{\Gamma(\alpha/2)}$ which corresponds to the Riesz kernel $\gamma(x) = |x|^{-\alpha}$. we need to show fractional noise is in weak L^2 in order to consider that. This already may be true but at the moment i am unsure. See nick question 1.8 in the github doc.. The reason for the restriction is the need to apply the Hardy-Littlewood-Sobolev lemma to show that ρ , defined below, is finite. This is shown in Lemma 1.6 in [5].

Remark 1.1. Let $c > 0$. The Riesz kernel $\gamma(x) = |x|^{-\alpha}$ and its spectral density $\varphi(x) = C_{d,\alpha}|x|^{-(d-\alpha)}$ satisfy the scaling properties $\gamma(cx) = c^{-\alpha}\gamma(x)$ and $\varphi(cx) = c^{-(d-\alpha)}\varphi(x)$. In addition, $\gamma = K * K$ where " $*$ " denotes the convolution and $K(x) = \beta_{\alpha,d}|x|^{-(d+\alpha)/2}$ with $\beta_{\alpha,d} = \pi^{-d/4} \frac{\Gamma((d+\alpha)/4)}{\Gamma((d-\alpha)/4)} \sqrt{\frac{\Gamma((d-\alpha)/2)}{\Gamma(\alpha/2)}}$.

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Remark 1.2. We may actually apply the techniques in this paper to any such noise whose correlation function and spectral density, γ and μ , satisfy

1. both μ and γ are absolutely continuous with respect to Lebesgue measure
2. for some $\alpha \in (0, d)$, γ satisfies that for all $c > 0$ and $x \in \mathbb{R}^d$ that $\gamma(cx) = c^{-\alpha}\gamma(x)$
3. there exists a nonnegative function K on $\mathbb{R}^d$ such that $\gamma = K * K$ where " $*$ " refers to the spatial convolution.

However, there is an issue with applying the results in this paper to the more general noises listed in 1.2. Proving ρ , defined below is finite is a challenge. Through inspection of the proof of Lemma 1.6 [5], we note that it suffices to show that

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \frac{F(\gamma)G(\lambda)}{\varphi(\lambda - \gamma)} d\gamma d\lambda \leq C \|F\|_r \|G\|_s \quad (1.3)$$

holds for some pair (r, s) where $F(x) = f(x)/(1+|x|^a)$ and G is defined similarly with f replaced by g and both f and g are $L^2(\mathbb{R}^d)$ functions with norm 1.

We mention that an example of a noise that differs from the Riesz kernel but satisfies 1.2 is the noise with $\gamma(x) = \prod_{i=1}^d |x_i|^{-\alpha_i}$ for some $\alpha_i \in (0, 1)$. The problem with this noise is that we are unsure if 1.3 holds. In this example, $\mu(d\xi) = \prod_{i=1}^d (C_{1,\alpha_i} |\xi_i|^{-(1-\alpha_i)} d\xi_i)$ and $K(x) = \prod_{i=1}^d (\beta_{\alpha_i,1} |x_i|^{-(1+\alpha_i)/2})$ and $\alpha = \sum_{i=1}^d \alpha_i$. If we consider the parameterization $\alpha_i = 2 - 2H_i$ with $H_i \in (1/2, 1)$, then the corresponding noise is the fractional Brownian sheet with indices $H_1, \dots, H_n$. The heat equation with this noise is studied in Hu [8]

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Here is the main result of this paper.

Theorem 1.3. *It holds that*

$$\lim_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p = \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta (\theta\rho)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \quad (1.4)$$

2 Some preliminaries.

2.1 Malliavin Calculus

Add Intro to Malliavin Calc Here

In general, we start with a non-negative and non-negative definite tempered measure Γ with density γ in the sense that $\Gamma(dx) = \gamma(x)dx$, and

$$\int_{\mathbb{R}^d} \Gamma(dx) (\varphi * \tilde{\varphi})(x) \geq 0 \quad \text{for all } \varphi \in \mathcal{S}(\mathbb{R}^d)$$

where $\varphi(\tilde{x}) = \varphi(-x)$. In addition we require the existence of an $r > 0$ such that

$$\int_{\mathbb{R}^d} \Gamma(dx) \frac{1}{(1 + |x|^2)^r} < \infty.$$

According to the Bochner-Schwarz theorem, there exists a non-negative measure μ , often referred to as the spectral measure on $\mathbb{R}^d$ whose Fourier transform is Γ in the sense that for any $\varphi \in (\mathbb{R}^d)$

$$\int_{\mathbb{R}^d} \Gamma(dx) \varphi(x) = \int_{\mathbb{R}^d} \mu(dt) \mathcal{F}\varphi(t)$$

The functional defined on the space of test functions $\mathcal{D}(\mathbb{R}^d)$

$$C(\varphi, \psi) = \int_{\mathbb{R}} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi)$$

is non-negative define and thus we may associate it as the covariance functional of a zero-mean Gaussian processes, $W := \{W(\varphi) : \varphi \in \mathcal{D}(\mathbb{R}^d)\}$. In other words,

$$\mathbb{E}(W(\varphi)W(\psi)) = \int_{\mathbb{R}} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi) =: \langle \varphi, \psi \rangle_{\mathcal{H}}.$$

Let $\mathcal{H}$ be the completion of $\mathcal{D}(\mathbb{R}^d)$ with respect to $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and thus we see $\varphi \rightarrow W(\varphi)$ is an isometry from the normed space $(\mathcal{D}(\mathbb{R}^d), \|\cdot\|_{\mathcal{H}})$ to $L^2(\Omega)$ where $\|f\|_{\mathcal{H}} = \langle f, f \rangle_{\mathcal{H}}$.

We denote δ the Skorohod integral with respect to W and we denote its domain $\text{Dom}\delta$. For any $u \in \text{Dom}\delta$, we write

$$\delta(u) = \int_{\mathbb{R}^d} u(x) W(\delta x).$$

Through construction of δ , $\delta(u)$ is a random variable in $L^2(\Omega)$. For a deeper explanation, we direct the reader to Nualart [9].

Definition 2.1. A random field $u = \{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$ such that $\|u(t, x)\|_2 < \infty$ for all $t \geq 0$ and $x \in \mathbb{R}^d$ is called a mild solution to the equation 1.1 if the random field $\{G(t - s, x - y)u(s, y) : s \in [0, t], y \in \mathbb{R}^d\}$ is Skorohod integrable for each $s \in [0, t]$ and $x \in \mathbb{R}^d$ fixed and satisfies the following integral equation:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t - s, x - y) u(s, y) W(\delta y) \right) ds \quad (2.1)$$

[Link the gap from the def of mild solution to piccard scheme comment and fill in the details.](#)

Through a Piccard Iteration Scheme we can see that the solution should be given by

$$u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)) \quad (2.2)$$

where

$$\begin{aligned} f_n(x_1, \dots, x_n; x, t) &= \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} G(t - t_n, x - x_n) \cdots G(t_2 - t_1, x_2 - x_1) dt_1 \cdots dt_n \\ &= \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} G(t_1, x - x_n) \cdots G(t_n - t_{n-1}, x_2 - x_1) dt_1 \cdots dt_n. \end{aligned} \quad (2.3)$$

For ease of notation, we write the above integrals as

$$f_n(x_1, \dots, x_n; x, t) = \int_{[0, t]_{<}^n} G(t - t_n, x - x_n) \cdots G(t_2 - t_1, x_2 - x_1) d\vec{t}$$

$$= \int_{[0,t]_{<}^n} G(t_1, x - x_n) \cdots G(t_n - t_{n-1}, x_2 - x_1) d\vec{t}.$$

We denote by $\tilde{f}_n(\cdot, x, t)$ to be the symmetrization of $f_n(\cdot, x, t)$ in the variables $x_1, \dots, x_n$:

$$\tilde{f}_n(\cdot; x, t) = \frac{1}{n!} \sum_{\rho \in \Sigma_n} f_n(x_{\rho(1)}, \dots, x_{\rho(n)}) \quad (2.4)$$

$$= \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{[0,t]_{<}^n} G(t - t_n, x - x_{\rho(n)}) \cdots G(t_2 - t_1, x_{\rho(2)} - x_{\rho(1)}) d\vec{t} \quad (2.5)$$

where Σ_n is the set of all permutations of $\{1, \dots, n\}$. In particular, for $x = 0$ we have

$$\tilde{f}_n(\cdot; 0, t) = \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{[0,t]_{<}^n} G(t_1, x_{\rho(1)}) \cdots G(t_n - t_{n-1}, x_{\rho(n)} - x_{\rho(n-1)}) d\vec{t}. \quad (2.6)$$

The Fourier transform of the kernels, f_n , is given by

$$\mathcal{F}f_n(\cdot; x, t)(\xi_1, \dots, \xi_n) = e^{-i(\sum_{j=1}^n \xi_j) \cdot x} \int_{[0,t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(t_{k+1} - t_k, \cdot)} \left(\sum_{j=1}^k \xi_j \right) dt \quad (2.7)$$

where $t_{n+1} = t$

Recall the notation above that $G(t, x) = Y_{a,b,r,d}(t, x)$. The following result shows an important scaling property that both $\mathcal{F}G(t, \cdot)(\xi)$ and $\|\tilde{f}_n(\cdot, t, x)\|_{\mathcal{H}^{\otimes n}}$ have.

Theorem 2.2. *Let $c > 0$, then the two following scaling properties hold:*

$$\mathcal{F}G(t, \cdot)(c\xi) = c^{-\frac{a}{b}(b+r-1)} \mathcal{F}G\left(c^{\frac{a}{b}}t, \cdot\right)(\xi) \quad \text{or} \quad \mathcal{F}G(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}G(t, \cdot)(c^{b/a}\xi) \quad (2.8)$$

and

$$\|\tilde{f}_n(\cdot, 0, t)\|_{H^{\otimes n}}^2 = t^{[2(b+r)-b\alpha/a]n} \|\tilde{f}_n(\cdot, 0, 1)\|_{H^{\otimes n}}^2 \quad (2.9)$$

Proof: The Fourier transform of G is given in [6]. With this we can directly calculate

$$\begin{aligned} \mathcal{F}G(t, \cdot)(c\xi) &= t^{b+r-1} \sum_{k=0}^{\infty} \frac{(-2^{-1}\nu t^b |c\xi|^a)^k}{\Gamma(bk + (b+r))} \\ &= t^{b+r-1} \sum_{k=0}^{\infty} \frac{(-2^{-1}\nu t^b c^a |\xi|^a)^k}{\Gamma(bk + (b+r))} \\ &= t^{b+r-1} \sum_{k=0}^{\infty} \frac{(-2^{-1}\nu (tc^{\frac{a}{b}})^b |\xi|^a)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= c^{\frac{a}{b}(1-r-b)} (tc^{\frac{a}{b}})^{b+r-1} \sum_{k=0}^{\infty} \frac{(-2^{-1}\nu (tc^{\frac{a}{b}})^b |\xi|^a)^k}{\Gamma(bk + (b + \gamma))} \\ &= c^{\frac{a}{b}(1-r-b)} \mathcal{F}G\left(c^{\frac{a}{b}}t, \cdot\right)(\xi) \end{aligned}$$

and this gives us the equality on 2.8. The second equality in 2.8 is proven by letting $c = c^{b/a}$ in the first equality.

For 2.9, first recall that

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 = \int_{\mathbb{R}^{nd}} |\mathcal{F} \tilde{f}_n(\cdot, 0, t)|^2(\vec{\xi}) \varphi(\vec{\xi}) d\vec{\xi}$$

where $\varphi(\vec{\xi}) = \varphi(\xi_1) \cdots \varphi(\xi_n)$. We start by applying this definition of $\tilde{f}_n$, 2.7 and the scaling property 2.9 to see

$$\begin{aligned} \int_{\mathbb{R}^{nd}} |\mathcal{F} \tilde{f}_n(\cdot, 0, t)|^2(\vec{\xi}) \varphi(\vec{\xi}) d\vec{\xi} &= \int_{\mathbb{R}^{nd}} \left| \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, t]^n <} \overline{\mathcal{F}G(t - t_n, \cdot) \left(\sum_{i=1}^n \xi_i \right) \cdots \mathcal{F}G(t_2 - t_1, \cdot) (\xi_1)} \right|^2 \varphi(\vec{\xi}) d\vec{\xi} \\ &= t^{2n(b+r-1)} \int_{\mathbb{R}^{nd}} \left| \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, t]^n <} \overline{\mathcal{F}G\left(1 - \frac{t_n}{t}, \cdot\right) (t^{b/a}(\sum_{i=1}^n \xi_i)) \cdots \mathcal{F}G\left(\frac{t_2}{t} - \frac{t_1}{t}, \cdot\right) (t^{b/a}\xi_1)} \right|^2 \varphi(\vec{\xi}) d\vec{\xi}. \end{aligned}$$

Now we apply the substitution $t'_k = t'_k/t$ which gives us

$$\begin{aligned} t^{2n(b+r-1)} \int_{\mathbb{R}^{nd}} \left| \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, t]^n <} \overline{\mathcal{F}G\left(1 - \frac{t_n}{t}, \cdot\right) (t^{b/a}(\sum_{i=1}^n \xi_i)) \cdots \mathcal{F}G\left(\frac{t_2}{t} - \frac{t_1}{t}, \cdot\right) (t^{b/a}\xi_1)} \right|^2 \varphi(\vec{\xi}) d\vec{\xi} \\ = t^{2n(b+r-1)+2n} \int_{\mathbb{R}^{nd}} \left| \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, 1]^n <} \overline{\mathcal{F}G(1 - t_n, \cdot) (t^{b/a}(\sum_{i=1}^n \xi_i)) \cdots \mathcal{F}G(t_2 - t_1, \cdot) (t^{b/a}\xi_1)} \right|^2 \varphi(\vec{\xi}) d\vec{\xi}. \end{aligned}$$

Next, we recall the scaling property of φ and apply the substitution $\xi'_k = t^{b/a}\xi_k$ which yields

$$\begin{aligned} t^{2n(b+r-1)+2n} \int_{\mathbb{R}^{nd}} \left| \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, 1]^n <} \overline{\mathcal{F}G(1 - t_n, \cdot) (t^{b/a}(\sum_{i=1}^n \xi_i)) \cdots \mathcal{F}G(t_2 - t_1, \cdot) (t^{b/a}\xi_1)} \right|^2 \varphi(\vec{\xi}) d\vec{\xi} \\ = t^{2n(b+r-1)+2n} \int_{\mathbb{R}^{nd}} \left| \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, 1]^n <} \overline{\mathcal{F}G(1 - t_n, \cdot) (\sum_{i=1}^n \xi_i) \cdots \mathcal{F}G(t_2 - t_1, \cdot) (\xi_1)} \right|^2 \varphi(t^{-b/a}\vec{\xi}) t^{-ndb/a} d\vec{\xi} \\ = t^{2n(b+r-1)+2n-nb\alpha/a} \int_{\mathbb{R}^{nd}} \left| \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \int_{[0, 1]^n <} \overline{\mathcal{F}G(1 - t_n, \cdot) (\sum_{i=1}^n \xi_i) \cdots \mathcal{F}G(t_2 - t_1, \cdot) (\xi_1)} \right|^2 \varphi(\vec{\xi}) d\vec{\xi} \\ = t^{2n(b+r-1)+2n-nb\alpha/a} \int_{\mathbb{R}^{nd}} |\mathcal{F} \tilde{f}_n(\cdot, 0, 1)|^2(\vec{\xi}) \varphi(\vec{\xi}) d\vec{\xi} \\ = t^{[2(b+r-1)+2-nb\alpha/a]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2 \\ = t^{[2(b+r)-nb\alpha/a]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2 \end{aligned}$$

□

Explain briefly the following standard procedure

The following result gives the existence and uniqueness of the solution. Its proof follows by a standard procedure.

Theorem 2.3. Fix any $T \in (0, \infty]$. Suppose that $f_n(\cdot, x, t) \in \mathcal{H}^{\otimes n}$ for any $t \in (0, T)$, $x \in \mathbb{R}^d$ and $n \geq 1$. Then equation 1.1 has a unique $L^2(\Omega)$ solution on $(0, T) \times \mathbb{R}^d$ if and only if the series 2.2 converges in $L^2(\Omega)$ for any $(t, x) \in (0, T) \times \mathbb{R}^d$, which is equivalent to the convergence of the series 2.10. In this case, the solution is given by 2.2 with the second moment given by

$$\mathbb{E}(|u(t, x)|^2) = \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \forall (t, x) \in (0, T) \times \mathbb{R}^d. \quad (2.10)$$

2.2 Some large deviation results.

We first define

$$\rho = \rho_{\nu,a} := \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{h(\xi+\eta)h(\eta)}{\sqrt{1+\frac{\nu}{2}|\xi+\eta|^a} \sqrt{1+\frac{\nu}{2}|\eta|^a}} d\eta \right]^2 \mu(d\xi). \quad (2.11)$$

where ν and a are from 1.1. In addition, $\mu(d\xi) = C_{d,\alpha}|\xi|^{-(d-\alpha)}$ which corresponds to the Riesz kernel $\gamma(x) = |x|^{-\alpha}$ for some $\alpha \in (0, d)$. In X. Chen [5], the case of $a \in (0, 2]$ and $\nu = 2$ is discussed and there it is proven that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \log[\rho_{2,a}]. \quad (2.12)$$

and in addition, X. Chen [5] also shows the equality $\rho_{2,a} = \Lambda_\alpha^{2-(\alpha/a)}$ where

$$\begin{aligned} \Lambda_\alpha &= \sup_{g \in \mathcal{F}_a} \left\{ \left(\int_{\mathbb{R}^d} \frac{g^2(x_1)g^2(x_2)}{|x_1+x_2|^a} d\vec{x} \right)^{1/2} - \mathcal{E}_a(g, g) \right\} \\ &= \sup_{g \in \mathcal{F}_a} \{ \langle g^2 * g^2, |\cdot|^{-a} \rangle - \mathcal{E}_a(g, g) \}. \end{aligned} \quad (2.13)$$

The following result shows the relationship between $\rho_{2,a}$ and ρ and well as the generalization of 2.12 for ρ .

Theorem 2.4. *For any $\nu > 0$ and $a \in (0, 2]$, we have that*

$$\rho = \left(\frac{\nu}{2} \right)^{-\alpha/a} \rho_{2,a} \quad (2.14)$$

and in addition,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \log[\rho]. \quad (2.15)$$

Proof: Recall that we define $\varphi(x) = C_{d,\alpha}|x|^{-(d-\alpha)}$. We begin by applying the substitution $x' = (\frac{\nu}{2})^{1/a}$ to ρ :

$$\begin{aligned} \rho &= \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a} \sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx) \\ &= \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \left(\frac{\nu}{2}\right)^{(d-\alpha)/a} \varphi(x) \left(\frac{\nu}{2}\right)^{-d/a} dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \varphi(x) dx \end{aligned}$$

Now define $f^*(x) = \left(\frac{\nu}{2}\right)^{\frac{-d}{2a}} f\left(\left(\frac{\nu}{2}\right)^{-1/a}x\right)$. Then with this we see that

$$\rho = \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y)f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx$$

and note that

$$\int_{\mathbb{R}^d} f(x)^2 dx = \int_{\mathbb{R}^d} f^*(x)^2 dx$$

and thus

$$\left\{ f \in L^2(\mathbb{R}^d) : \|f\|_2 = 1 \right\} = \left\{ f \in L^2(\mathbb{R}^d) : \|f^*\|_2 = 1 \right\}$$

so we can say that

$$\begin{aligned} \rho &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y) f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f^*\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y) f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \rho_{2,a} \end{aligned}$$

and this proves 2.14.

To prove 2.15, we first notice that by applying the substitution $\xi'_{\sigma(j)} = (\frac{\nu}{2})^{1/a} \xi_{\sigma(j)}$ that

$$\int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}) = \left(\frac{\nu}{2}\right)^{-n\alpha/a} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}) \quad (2.16)$$

and now we can directly calculate

$$\begin{aligned} &\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right)^2 \mu(d\vec{\xi}) \right] \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \left(\frac{\nu}{2}\right)^{-n\alpha/a} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}) \right] \\ &= \log \left[\left(\frac{\nu}{2}\right)^{-\alpha/a} \right] + \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}) \right] \end{aligned}$$

and because of both Theorem 1.5 [5] and 2.14 above,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right)^2 \mu(d\vec{\xi}) \right] = \log \left[\left(\frac{\nu}{2}\right)^{-\alpha/a} \right] + \log[\rho_{2,a}] = \log[\rho]$$

□

I never mentioned $\mathcal{M}$. Since it is essentially the same thing as Λ I may leave it out

3 Solvability

The goal for this section is to prove the following theorem:

Theorem 3.1. *The equation 1.1 has a unique solution $\{u(t, x) : t > 0, x \in \mathbb{R}^d\}$ and*

$$\sup_{(t,x) \in [0,T] \times \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) < \infty \quad \text{for all } p \geq 2, T > 0 \quad (3.1)$$

We need to build up some theory before we prove the above theorem. We first introduce some notation that will be used throughout the remainder of his paper. We start by defining $L_n(\vec{y})$ to be the Laplace transform of $\tilde{f}_n(\vec{y}, 0, \cdot)$ evaluated at 1:

$$\begin{aligned} L_n(\vec{y}) &= n! \int_0^\infty e^{-t} \tilde{f}_n(\vec{y}; 0, t) dt \\ &= \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0, t]^n} \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{s} dt \end{aligned}$$

with the convention that. $s_0 = y_{\sigma(0)} = 0$. Next we define,

$$\begin{aligned} H_n(t, \vec{x}) &= n! \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \tilde{f}_n(\vec{y}; 0, t) d\vec{y} \\ &= \sum_{\sigma \in \Sigma_n} \int_{[0, t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{y} ds \end{aligned}$$

where in the last line we used relation 2.6 for expressing $\tilde{f}_n(\cdot; 0, t)$. Next, similarly to L_n , we define $L(x)$ and can calculate $\mathcal{F}L(\xi)$ to be

$$L(x) = \int_0^\infty e^{-t} G(t, x) dt \quad \text{and} \quad \mathcal{F}L(\xi) = \int_0^\infty e^{-t} \mathcal{F}G(t, \cdot)(\xi) dt = \frac{1}{1 + \frac{\nu}{2} |\xi|^a} \quad (3.2)$$

and through a change of variables, one sees that the Laplace transform of convolution in the time variable become a product:

$$L_n(\vec{y}) = \sum_{\sigma \in \Sigma_n} L(y_{\sigma(1)}) L(y_{\sigma(2)} - y_{\sigma(1)}) \cdots L(y_{\sigma(n)} - y_{\sigma(n-1)}) \quad (3.3)$$

and using this and 3.2, we can calculate $\mathcal{F}L_n(\vec{\xi})$:

$$\mathcal{F}L_n(\vec{\xi}) = \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a} \quad (3.4)$$

Lemma 3.2. *Insert text about assumptions*

$$\int_{(\mathbb{R}^d)^n} \left[\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right]^2 d\vec{x} = \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_n(\vec{\xi})|^2 \mu(d\vec{\xi}) \quad (3.5)$$

Proof: Same proof from wave paper. Copy proof there

Lemma 3.3. *Insert text about assumptions*

$$\left\| \tilde{f}_n(\cdot; 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_n^2(t, \vec{x}) d\vec{x} \quad (3.6)$$

Proof: Same proof from wave paper. Copy proof later.

If f is a non-decreasing function on $[0, \infty)$, then the following reverse Cauchy-Schwarz inequality holds, which plays a key role in our developments.

Lemma 3.4. *If $f : [0, \infty) \rightarrow [0, \infty)$ is a non-decreasing function, then*

$$2 \int_0^\infty e^{-2t} f^2(t) dt \leq \left(\int_0^\infty e^{-t} f(t) dt \right)^2 \quad (3.7)$$

Proof: Let τ and τ' be two independent random variables with exponential distribution of mean 1. Then $\tau \wedge \tau'$ has an exponential distribution with density $g(t) = 2e^{-2t}$ and

$$\left(\int_0^\infty e^{-t} f(t) dt \right)^2 = \mathbb{E}[f(\tau)] \mathbb{E}[f(\tau')] = \mathbb{E}[f(\tau)f(\tau')] = 2 \int_0^\infty e^{-2t} f^2(t) dt$$

□

Denote

$$T_n = T_n(\nu, a) := \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}). \quad (3.8)$$

By applying the substitution $\xi'_k = (\frac{\nu}{2})^{1/a} \xi_k$ and the scaling property for φ , we may rewrite T_n as

$$T_n = \left(\frac{\nu}{2} \right)^{-n\alpha/a} T_n(2, a) \quad (3.9)$$

Lemma 3.5. *Under the assumption of Theorem [main theorem on the moment asymptotics that I need to put into the introduction](#), we have that*

$$C_\mu = C_\mu(\nu, a) := \int_{(\mathbb{R}^d)} \left(\frac{1}{1 + \frac{\nu}{2} |\xi|^a} \right)^2 \mu(d\xi) < \infty \quad \text{and} \quad T_n \leq (n!)^2 (C_\mu)^2 \quad (3.10)$$

Proof: By the same spherical substitution in the wave paper we see that

$$\int_{(\mathbb{R}^d)} \left(\frac{1}{1 + |\xi|^a} \right)^2 \mu(d\xi) = \left(\int_{S^{d-1}} \varphi(\gamma) d\sigma(\gamma) \right) \left(\int_0^\infty \frac{\rho^{\alpha-1}}{(1 + \rho^a)^2} d\rho \right)$$

The first integral is shown to be finite in the wave paper. The second integral is finite if

$$2a - \alpha + 1 > 1 \quad (3.11)$$

The same arguments from the wave paper show that

$$\begin{aligned} T_n &\leq n! \int_{(\mathbb{R}^d)^n} \sum_{\sigma \in \Sigma_n} \left[\prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\xi_1) \cdots \mu(d\xi_n) \\ &= (n!)^2 \int_{(\mathbb{R}^d)^n} \left[\prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a} \right]^2 \mu(d\xi_1) \cdots \mu(d\xi_n) \\ &\leq (n!)^2 \left[\sup_{\eta \in \mathbb{R}^d} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left(\frac{1}{1 + \frac{\nu}{2} |\xi + \eta|^a} \right)^2 \mu(d\xi) \right]^n \\ &= (n!)^2 \left[\int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left(\frac{1}{1 + \frac{\nu}{2} |\xi|^a} \right)^2 \mu(d\xi) \right]^n \\ &= (n!)^2 (C_\mu)^n \end{aligned}$$

where the last line follows by Lemma 4.1 of Balan and Song [2] or by Lemma 3.1 in [4]. □

Lemma 3.6. *The following inequality shows that $\tilde{f}_n(\cdot, 0; t) \in \mathcal{H}_{\otimes n}^2$ for any $t > 0$.*

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}_{\otimes n}}^n dt \leq \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]} T_n}{(n!)^2} \leq 2^{n[2(b+r) - \frac{b\alpha}{a}]} (C'_\mu)^n$$

Proof: We first derive the following scaling property for $\mathcal{F}\tilde{f}_n$ where $\tilde{f}_n$ is the kernel of the n^{th} Wiener integral in the chaos expansion of the solution.

$$\mathcal{F}\tilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = c^{n(b+r)} \mathcal{F}\tilde{f}_n(\cdot, 0, t)(c^{b/a}\xi_1, \dots, c^{b/a}\xi_n) \quad (3.12)$$

Let $c > 0$.

$$\mathcal{F}\tilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \mathcal{F}f_n(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)})$$

so it suffices to just look at the $\mathcal{F}f(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)})$ terms individually.

$$\begin{aligned} \mathcal{F}f_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) &= \int_{[0, ct]^n} \prod_{k=1}^n \overline{\mathcal{F}G(t_{k+1} - t_k, \cdot)} \left(\sum_{j=1}^k \xi_j \right) d\vec{t} \\ &= \int_{[0, t]^n} \prod_{k=1}^n \overline{\mathcal{F}G(c(t_{k+1} - t_k), \cdot)} \left(\sum_{j=1}^k \xi_j \right) c^n d\vec{t} \end{aligned}$$

where we applied the substitution $t'_i = t_i/c$ by applying the scaling property of $\mathcal{F}G$ we see that

$$\begin{aligned} \int_{[0, t]^n} \prod_{k=1}^n \overline{\mathcal{F}G(c(t_{k+1} - t_k), \cdot)} \left(\sum_{j=1}^k \xi_j \right) c^n d\vec{t} &= \int_{[0, t]^n} \prod_{k=1}^n \overline{c^{b+r-1} \mathcal{F}G(t_{k+1} - t_k, \cdot)} \left(c^{b/a} \sum_{j=1}^k \xi_j \right) c^n d\vec{t} \\ &= c^{n(b+r)} \mathcal{F}f_n(\cdot, 0, t)(c^{b/a}\xi_1, \dots, c^{b/a}\xi_n) \end{aligned}$$

Now by using this scaling property we see that

$$\begin{aligned} \int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \int_0^\infty e^{-t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\tilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\ &= \int_0^\infty e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\tilde{f}_n(\cdot, 0; 2t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) 2dt \\ &= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\tilde{f}_n(\cdot, 0, t)(2^{b/a}\xi_1, \dots, 2^{b/a}\xi_n) \right|^2 \mu(d\vec{\xi}) dt \\ &= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\tilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 2^{-\frac{nb\alpha}{a}} 2^{\frac{nb(d-\alpha)}{a}} \mu(d\vec{\xi}) dt \\ &= 2^{n(2(b+r)-b\alpha/a)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F}\tilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\ &= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt \end{aligned}$$

and now we apply Lemmas 3.3 and 3.4 as well as Equation 3.4 to get

$$\begin{aligned} \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt &\leq \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_{\mathbb{R}^{nd}} \left(\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right)^2 d\vec{x} \\ &= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_{\mathbb{R}^{nd}} |\mathcal{F}L_n(\vec{\xi})|^2 d\vec{\xi} \\ &= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} T_n \end{aligned}$$

and this proves the first inequality. An application of Lemma 3.6 gives is the second inequality. $\square$

Proof of Theorem 3.1: First recall 3.12, which says

$$\mathcal{F}\tilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = c^{n(b+r)} \mathcal{F}\tilde{f}_n(\cdot, 0, t)(c^{b/a}\xi_1, \dots, c^{b/a}\xi_n).$$

In order to prove the theorem we need to show that $\sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 < \infty$. From the scaling property 2.9 we see that

$$\begin{aligned} \int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \int_0^\infty e^{-t} t^{[2(b+r) - \frac{b\alpha}{a}]n} dt \\ &= \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \end{aligned}$$

as long as we have

$$2(b+r) - \frac{b\alpha}{a} > 0 \quad (3.13)$$

By using Lemma 3.6 we can deduce that

$$\begin{aligned} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 &= \frac{\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \\ &\leq \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]} T_n}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \\ &\leq \frac{\left(2^{[2(b+r) - \frac{b\alpha}{a}]} C'_\mu \right)^n}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \end{aligned}$$

By the same reasoning as in the proof in the wave paper (details in my notebook) we can say that there exists a constant depending on α, k_α , such that

$$\left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \leq k_\alpha \frac{\left(2^{[2(b+r) - \frac{b\alpha}{a}]} C'_\mu \right)^n}{(n!)^{2(b+r) - b\alpha/a}}$$

Therefore,

$$\begin{aligned} \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 &\leq \sum_{n \geq 0} \theta^n n! k_\alpha \frac{\left(2^{[2(b+r) - \frac{b\alpha}{a}]} C'_\mu \right)^n}{(n!)^{2(b+r) - b\alpha/a}} \\ &= \sum_{n \geq 0} \theta^n k_\alpha \frac{\left(2^{[2(b+r) - \frac{b\alpha}{a}]} C'_\mu \right)^n}{(n!)^{2(b+r) - 1 - b\alpha/a}} \end{aligned}$$

and the last sum is finite when

$$2(b+r) - 1 - b\alpha/a > 0 \quad (3.14)$$

$\square$

4 The case $p = 2$.

In this section, we will prove the following result.

Theorem 4.1.

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) = \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta (\theta\rho)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right)$$

In order to prove this theorem, we first need to prove the following lemma.

Lemma 4.2.

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right) = \log \left(2^{2(b+r) - \frac{b\alpha}{a}} \right) + \log[\rho] \quad (4.1)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{2(b+r) - \frac{b\alpha}{a}} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[\left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \right] + \log[\rho] \quad (4.2)$$

Proof: Recall the definition of T_n from above as well as 2.15 which says that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{T_n}{(n!)^2} \right] = \log[\rho].$$

We already showed in the proof of Theorem 3.1 above that

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)$$

and then Lemma 3.6 above shows that

$$\left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \leq \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]} T_n}{(n!)^2}$$

Combining this and 2.15 we see that

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{1}{n} \log \left[\left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right] \\ \leq \log \left(2^{[2(b+r) - \frac{b\alpha}{a}]} \right) + \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{T_n}{(n!)^2} \right) \\ = \log \left(2^{[2(b+r) - \frac{b\alpha}{a}]} \right) + \log(\rho) \end{aligned}$$

which proves the upper bound for 4.1.

Now we prove the lower bound. Define for $t > 0$ and $t' > 0$

$$J_n(t, t') = \int_{\mathbb{R}^{nd}} H_n(t, x) H_n(t', x) dx$$

and recall Lemma 3.3 which says

$$\left\| \tilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_n(t, \vec{x})^2 d\vec{x} = \frac{1}{(n!)^2} J_n(t, t).$$

Thus $J_n(t, t)$ satisfies the scaling property

$$J_n(t, t) = t^{[2(b+r)-b\alpha/a]n} J_n(1, 1).$$

Let τ and $\tilde{\tau}$ be independent exponential random variables with mean 1. By lemma 3.2,

$$\begin{aligned} \mathbb{E}[J_n(\tau, \tau')] &= \int_0^\infty \int_0^\infty e^{-t} e^{-\tilde{t}} J_n(t, \tilde{t}) dt d\tilde{t} = \int_{\mathbb{R}^{dn}} \left[\int_0^\infty e^{-t} H_n(t, x) dt \right]^2 dx \\ &= \int_{\mathbb{R}^{nd}} |\mathcal{F}L_n(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \cdots \mu(d\xi_n) = T_n \end{aligned}$$

In addition,

$$J_n(t, t') \leq J_n(t, t)^{1/2} J_n(t', t')^{1/2} = t^{[2(b+r)-b\alpha/a]n/2} t'^{[2(b+r)-b\alpha/a]n/2} J_n(1, 1).$$

Combining all of this gives us

$$\begin{aligned} T_n &= \mathbb{E}[J_n(\tau, \tilde{\tau})] \leq \mathbb{E} \left[\tau^{[2(b+r)-b\alpha/a]n/2} \right] \mathbb{E} \left[\tilde{\tau}^{[2(b+r)-b\alpha/a]n/2} \right] J_n(1, 1) \\ &= \Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 J_n(1, 1) \\ &= \Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 (n!)^2 \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \end{aligned}$$

where the last line follows from Lemma 3.3. Thus so far we have shown

$$\frac{T_n}{(n!)^2} = \frac{\left(\frac{\nu}{2}\right)^{-n\alpha/a} T'_n}{(n!)^2} \leq \Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad (4.3)$$

Now, another application of Stirling's formula shows that

$$\Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 \sim \Gamma([2(b+r)-b\alpha/a]n+1) 2^{-[2(b+r)-b\alpha/a]n} C_n$$

where $C_n = 2^{-1}[2(b+r)-b\alpha/a]^{1/2}(2\pi n)^{1/2}$. We now infer from 4.3 that

$$c_\alpha \frac{2^{[2(b+r)-b\alpha/a]n}}{C_n} \frac{1}{(n!)^2} T_n \leq \Gamma([2(b+r)-b\alpha/a]n+1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2$$

where c_α is a constant depending on α . Since $\frac{1}{n} \log(C_n) \rightarrow 0$ as $n \rightarrow \infty$, we obtain the lower bound

$$\begin{aligned} \liminf_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma([2(b+r)-b\alpha/a]n+1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) \\ \geq \log \left(2^{[2(b+r)-b\alpha/a]} \right) + \log(\rho) \end{aligned}$$

Thus 4.1 is proven. Lastly, 4.2 follows from 4.1 and the fact that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{\Gamma(an+1)}{(n!)^a} \right] = a \log(a).$$

Maybe fill in the details for proving 4.2 by using 4.1 and the above fact □

With the above lemma, we can now prove Theorem 4.1.

Proof of Theorem 4.1: Assuming that there is an $L^2(\Omega)$ solution $u(t, x)$ and then applying the scaling property 2.9 gives us

$$\begin{aligned}\mathbb{E}(|u(t, x)|^2) &= \sum_{n \geq 0} \theta^n (n!) \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \sum_{n \geq 0} \theta^n (n!) t^{(2(b+r)-b\alpha/a)n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} x_n R_n t^{(2(b+r)-b\alpha/a)n}\end{aligned}$$

where

$$R_n = (n!)^{2(b+r)-b\alpha/a} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \text{and} \quad x_n = \frac{\theta^n}{(n!)^{2(b+r)-(b\alpha/a)-1}}$$

4.2 above says that

$$\frac{1}{n} \log(R_n) \rightarrow \log \left[\left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho \right] \quad \text{as } n \rightarrow \infty.$$

Now define R to be

$$R = \left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho.$$

We want to find a β and A so that

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \sum_{n \geq 0} x_n R^n \left(t^{(2(b+r)-b\alpha/a)} \right)^n = A$$

By lemma A3 notice that

$$\lim_{t \rightarrow \infty} \left[\frac{1}{(\theta R) t^{2(b+r)-b\alpha/a}} \right]^{\frac{1}{2(b+r)-(b\alpha/a)-1}} \log \sum_{n \geq 0} \frac{[(\theta R) t^{2(b+r)-b\alpha/a}]^n}{(n!)^{2(b+r)-(b\alpha/a)-1}} = 2(b+r) - (b\alpha/a) - 1$$

and by an easy algebraic manipulation

$$\begin{aligned}\lim_{t \rightarrow \infty} \left[\frac{1}{t^{2(b+r)-b\alpha/a}} \right]^{\frac{1}{2(b+r)-(b\alpha/a)-1}} \log \sum_{n \geq 0} \frac{[(\theta R) t^{2(b+r)-b\alpha/a}]^n}{(n!)^{2(b+r)-(b\alpha/a)-1}} \\ = [2(b+r) - (b\alpha/a) - 1] (\theta R)^{\frac{1}{2(b+r)-(b\alpha/a)-1}}\end{aligned}$$

hence we see that

$$\beta = \frac{2(b+r) - b\alpha/a}{2(b+r) - (b\alpha/a) - 1} \quad \text{and} \quad A = [2(b+r) - (b\alpha/a) - 1] (\theta R)^{\frac{1}{2(b+r)-(b\alpha/a)-1}}$$

or by removing the fractions in the denominator

$$\beta = \frac{2a(b+r) - b\alpha}{2a(b+r) - (b\alpha) - a} \quad \text{and} \quad A = [2(b+r) - (b\alpha/a) - 1] (\theta R)^{\frac{a}{2a(b+r)-b\alpha-a}}$$

Finally, lemma A2 says that

$$\begin{aligned}\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) \\ = \theta^{\frac{a}{2a(b+r)-b\alpha-a}} \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \rho^{\frac{a}{2a(b+r)-b\alpha-a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right)\end{aligned} \quad (4.4)$$

5 Upper Bound.

insert intro to section

Theorem 5.1.

$$\limsup_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \leq \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta (\theta\rho)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right) \quad (5.1)$$

Proof: We denote u_λ for $\lambda > 0$ to be the the solution of the fractional equation with θ replaced with λ and $u = u_\theta$. Using theorem B1 from the wave paper we can say that for any $p \geq 2$ that

$$\|u(t, 0)\|_p \leq \|u_{(p-1)\theta}(t, 0)\|_2$$

and by the scaling property

$$\begin{aligned} \|u_{(p-1)\theta}(t, 0)\|_2^2 &= \sum_{n \geq 0} (n!)^n \theta^n (p-1)^n \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} (n!)^n \theta^n \left\| \tilde{f}_n\left(\cdot, 0, t(p-1)^{\frac{1}{2(b+r) - b\alpha/a}}\right) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \left\| u\left(t(p-1)^{\frac{1}{2(b+r) - b\alpha/a}}, 0\right) \right\|_2^2 \end{aligned}$$

and from this we can see that

$$\|u(t, 0)\|_p \leq \left\| u\left(t(p-1)^{\frac{1}{2(b+r) - b\alpha/a}}, 0\right) \right\|_2 =: \|u(t_p, 0)\|_2$$

and an application of Theorem 4.1 proves the result. □

6 Lower Bound.

The goal of this section is to prove the following theorem.

Theorem 6.1.

$$\liminf_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \geq \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta (\theta\rho)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right)$$

We will need to build up some theory before being able to prove the above theorem. We start by defining the function $W_n(t, \phi)$ on $(0, \infty) \times L_{\mathbb{C}}^2(\mu)$ by

$$W_n(t, \phi) := \int_{[0, t]^n <} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \mathcal{F}G(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) \mu(d\xi_1) \dots \mu(d\xi_n) ds \quad (6.1)$$

with $s_0 = 0$. We now give conditions for when W_n is well defined.

Lemma 6.2.

$$\int_0^\infty e^{-t} W_n(t, \phi) dt = \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi) \dots \mu(d\xi_n)$$

Proof: same as wave paper. copy later.

Proposition 6.3. For $f \in \mathcal{H}$, $t > 0$, and $p, q > 1$ with $p^{-1} + q^{-1} = 1$, it holds that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2}(q-1) \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f) \right|$$

Proof: same as wave paper. copy later.

Our next goal is to prove the following proposition, which is where the motivation of the definitions of t_p and β come from.

Proposition 6.4. For $p, q > 1$ and $p^{-1} + q^{-1} = 1$ we have that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \right|$$

Proof: The above inequality motivates us to find a function $f_\tau(x)$ and some τ such that

1.

$$(q-1) \|f_\tau\|_{\mathcal{H}}^2 = (p-1)^{-1} \|f_\tau\|_{\mathcal{H}}^2 = t_p^\beta \|f\|_{\mathcal{H}}^2$$

2.

$$W_n(t, \mathcal{F}f_\tau) = W_n(t_p^\beta, \mathcal{F}f)$$

since then we could just apply Proposition 6.3 to the function f_τ . Note above that we have not yet defined β or t_p . We guess f_τ takes the form $f_\tau(x) = \tau^V f(\tau^W x)$. We now mention, through direct calculation one can show that $\|f_\tau\|_{\mathcal{H}}^2$ and $W_n(t, \mathcal{F}f_\tau)$ can be calculated as follows:

$$\|f_\tau\|_{\mathcal{H}}^2 = \tau^{2(V-dW)+W\alpha} \|f\|_{\mathcal{H}}^2 \quad (6.2)$$

and

$$W_n(t, \mathcal{F}f_\tau) = \tau^n [V - W((d-\alpha) + \frac{a}{b}(b+r))] W_n(t\tau^{\frac{a}{b}W}, \mathcal{F}f). \quad (6.3)$$

This means we must have

$$\begin{cases} V - W((d-\alpha) + \frac{a}{b}(b+r)) &= 0 \\ (p-1)^{-1} \tau^{2(V-dW)+W\alpha} &= t\tau^{\frac{a}{b}W} \end{cases}$$

We need to now choose an appropriate τ and we choose $\tau = (p-1)t$ and so the above system of equations becomes

$$\begin{cases} V - W((d-\alpha) + \frac{a}{b}(b+r)) &= 0 \\ (p-1)^{2(V-dW)+W\alpha-1} t^{2(V-dW)+W\alpha} &= (p-1)^{\frac{a}{b}W} t^{\frac{a}{b}W+1} \end{cases}$$

Equating the exponents of either the $(p-1)$ term or the t term both lead to the following equation

$$2(V-dW) + W\alpha - 1 - \frac{a}{b}W = 0 \quad (6.4)$$

and so now we have two linear equations in W and V that we can solve, namely

$$\begin{cases} V - W((d-\alpha) + \frac{a}{b}(b+r)) &= 0 \\ 2(V-dW) + W\alpha - 1 - \frac{a}{b}W &= 0 \end{cases}$$

Now if we define

$$\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1} \quad (6.5)$$

then solving the system gives us

$$\begin{cases} W = \frac{b}{a}(\beta - 1) \\ V = \left(\frac{a}{b}(b+r) - \alpha + d\right) \frac{b}{a}(\beta - 1) \end{cases}.$$

Lastly if we define

$$t_p^\beta := (q-1)(p-1)^\beta t^\beta = (p-1)^{\beta-1} t^\beta \quad (6.6)$$

then Proposition 6.4 is proven by applying Proposition 6.3 to the function

$$f_\tau(x) = \tau^{\left(\frac{a}{b}(b+r) - \alpha + d\right) \frac{b}{a}(\beta-1)} f\left(\tau^{\frac{b}{a}(\beta-1)} x\right)$$

with $\tau = (p-1)t$.

□

We now want to identify all the $f \in \mathcal{H}$ for which $W_n(t, \mathcal{F}f)$ is nonnegative. In fact, if we consider the space

$$\mathcal{H}_+ = \{f \in \mathcal{H} : f \text{ is nonnegative and nonnegative definite}\}$$

then by Plancherel's theorem

$$W_n(t, \mathcal{F}(f)) = \int_{[0,t]^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n (f * \gamma)(x_k) \prod_{k=1}^n G(s_k - s_{k-1}, x_k - x_{k-1}) dx ds =: U_n(t, f)$$

with $s_0 = 0$ and $x_0 = 0$. Under assumptions on the parameters, the Green's function for the fractional equation is nonnegative and hence under these conditions we get that $W_n(t, \mathcal{F}(f)) \geq 0$ for all the functions $f \in \mathcal{H}_+$.

Now we define

$$W_n(t) := \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} W_n(t, \mathcal{F}(f)) \quad \text{and} \quad U_n(t) := \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} U_n(t, f)$$

and with this we see that

$$W_n(t) \geq U_n(t) \geq 0$$

Theorem 6.5. *If τ is an exponential random variable with mean 1, then*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(\tau)) \geq \log(\rho^{1/2})$$

Proof: We start by letting τ be an exponential random variable with mean 1. With this,

$$\mathbb{E}(U_n(\tau)) = \int_0^\infty e^{-t} U_n(t) dt \geq \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \int_0^\infty e^{-t} U_n(t, f) dt$$

For an $f \in \mathcal{H}_+$ and $\|f\|_{\mathcal{H}} = 1$ we have that $\mathcal{F}(f)$ is even and in addition, Lemma 6.2 gives us that

$$\int_0^\infty e^{-t} U_n(t, f) dt = \int_0^\infty e^{-t} W_n(t, \mathcal{F}f) dt = \int_{\mathbb{R}^{dn}} \prod_{k=1}^n \mathcal{F}f(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \dots \mu(d\xi_n)$$

Modifying the ideas in the Xia Chen paper, we can say that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \int_{\mathbb{R}^{dn}} \prod_{k=1}^n (\mathcal{F}f)(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \cdots \mu(d\xi_n) = \log[\rho(\mathcal{F}f)]$$

where

$$\rho(g) := \sup_{\|h\|_{L^2}=1} \int_{\mathbb{R}^d} g(\xi) \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2} |\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2} |\eta|^a}} d\eta \right] \mu(d\xi)$$

and with this being said, we see that for any $f \in \mathcal{H}_+$ and $\|f\|_H = 1$

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(t)) \geq \log \rho(\mathcal{F}f)$$

We now need to calculate

$$\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f)$$

Consider a function $h \in L^2(\mathbb{R}^d)$. The function $h(\cdot)/\sqrt{1 + \frac{\nu}{2} |\cdot|^a} \in L^2(\mathbb{R}^d)$ so that $g(x) := (2\pi)^{d/2} \mathcal{F}^{-1} \left(\frac{h(\cdot)}{\sqrt{1 + \frac{\nu}{2} |\cdot|^a}} \right) (x)$ is well defined. **I still need to verify the details that** under these conditions, $g \in W_{1,a}(\mathbb{R}^d)$ with $\|g\|_{W_{1,a}} = 1$. Here is a sketch of the details

$$\begin{aligned} \|g\|_{W_{1,a}}^2 &= \|g\|_{L^2}^2 + (2\pi)^{-d} \sum_{k=1}^d \int_{\mathbb{R}^d} \xi_k^a |\mathcal{F}g(\xi)|^2 d\xi \\ &= (2\pi)^{-d} \int_{\mathbb{R}^d} (1 + |\xi|^a) |\mathcal{F}g(\xi)|^2 d\xi \\ &= \int_{\mathbb{R}^d} h(\xi)^2 d\xi = 1 \end{aligned}$$

where above we used the fact that $\|g\|_{L^2}^2 = (2\pi)^{-d} \|\mathcal{F}g\|_{L^2}^2$.

Now, for notation let $\tilde{h}(x) = h(-x)$. Note that

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2} |\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2} |\eta|^a}} d\eta &= (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}g(\xi - \eta) \mathcal{F}g(-\eta) d\eta \\ &= (2\pi)^{-d} (\mathcal{F}g * \widetilde{\mathcal{F}g})(\xi) \end{aligned}$$

Since $h(\cdot)$ is real value then $\widetilde{\mathcal{F}g} = \overline{\mathcal{F}g} = \mathcal{F}\bar{g}$. Using this and the fact that $\mathcal{F}(fg) = (2\pi)^{-d} \mathcal{F}(f) * \mathcal{F}(g)$, we see that $(2\pi)^{-d} (\mathcal{F}g * \widetilde{\mathcal{F}g})(\xi) = \mathcal{F}[|g|^2](\xi)$ and with this we see that

$$\begin{aligned} \rho(\mathcal{F}f) &= \sup_{\|h\|_{L^2}=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) (\mathcal{F}g * \widetilde{\mathcal{F}g})(\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2}=1} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) \mathcal{F}[|g|^2](\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2}=1} \langle f, |g|^2 \rangle_{\mathcal{H}}. \end{aligned}$$

With this we see that

$$\sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} \sup_{\|h\|_{L^2}=1} \langle f, |g|^2 \rangle_{\mathcal{H}} = \sup_{\|h\|_{L^2}=1} \langle |g|^2, |g|^2 \rangle_{\mathcal{H}}^{1/2}$$

where the optimal f is chosen to be $|g|^2 / \| |g|^2 \|_{\mathcal{H}}$. **Can the optimal f be chosen like this? This is true for dx but is it true for $\mu(dx)$? I believe that it would be true for μ such that μ has a density which is radially symmetric about the origin. The Riesz kernel would then still be ok** Therefore the supremum can be restricted to $\mathcal{H}_+$ which implies

$$\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = \sup_{\|h\|_{L^2}=1} \langle |g|^2, |g|^2 \rangle_{\mathcal{H}}^{1/2}$$

and notice that

$$\begin{aligned} \langle |g|^2, |g|^2 \rangle_{\mathcal{H}}^{1/2} &= \left(\int_{\mathbb{R}^d} |\mathcal{F}[|g|^2](\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= (2\pi)^{-d} \left(\int_{\mathbb{R}^d} |(\mathcal{F}g * \tilde{\mathcal{F}}g)(\xi)|^2 \mu(d\xi) \right)^{1/2} \\ &= \left(\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2}|\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2}|\eta|^a}} d\eta \right]^2 \mu(d\xi) \right)^{1/2} \\ &= \rho^{1/2} \end{aligned}$$

which finally leads us to the equality

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(\tau)) \geq \log(\rho^{1/2})$$

□

We now give scaling properties for W_n and U_n .

Lemma 6.6. *We have that*

$$W_n(t) = t^{n \frac{b}{a} (a - \frac{\alpha}{2} + \frac{a}{b} r)} W_n(1)$$

and moreover

$$U_n(t) = t^{n \frac{b}{a} (a - \frac{\alpha}{2} + \frac{a}{b} r)} U_n(1)$$

Proof: Let $\phi = \mathcal{F}f$ and $c > 0$ There may be a typo in this work, I think on the third line there should no longer be a c inside $\varphi(c\xi)$. Ill check later.

$$\begin{aligned} W_n(t, \phi(c^{-1} \cdot)) &= \int_{[0, t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(c^{-1} \xi_k) \prod_{k=1}^n G(s_k - s_{k-1}, \cdot) (\xi_k + \cdots \xi_n) \prod_{i=1}^n \varphi(\xi_i) d\xi_i ds_i \\ &= c^{nd} \int_{[0, t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n G(s_k - s_{k-1}, \cdot) (c(\xi_k + \cdots \xi_n)) \prod_{i=1}^n \varphi(c\xi_i) d\xi_i ds_i \\ &= c^{nd} c^{-n(d-\alpha)} c^{-n \frac{a}{b} (b+r-1)} \int_{[0, t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n G\left(c^{\frac{b}{a}}(s_k - s_{k-1}), \cdot\right) (\xi_k + \cdots \xi_n) \prod_{i=1}^n \varphi(c\xi_i) d\xi_i ds_i \\ &= c^{nd} c^{-n(d-\alpha)} c^{-n \frac{a}{b} (b+r-1)} c^{-n \frac{a}{b}} \int_{[0, c^{\frac{a}{b}} t]_{<}^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n G(s_k - s_{k-1}, \cdot) (\xi_k + \cdots \xi_n) \prod_{i=1}^n \varphi(c\xi_i) d\xi_i ds_i \\ &= c^{-n(a-\alpha+\frac{a}{b}r)} W_n\left(c^{\frac{a}{b}} t, \phi\right) \end{aligned}$$

Now we apply this with $c^{\frac{a}{b}} = \frac{1}{t}$ to get

$$W_n\left(t, \phi\left(t^{\frac{b}{a}} \cdot\right)\right) = t^{n \frac{b}{a} (a - \alpha + \frac{a}{b} r)} W_n(1, \phi)$$

Now note that

$$\left\| \phi\left(t^{\frac{b}{a}} \cdot\right) \right\|_{L^2(\mu)}^2 = t^{-\frac{\alpha b}{a}} \|\phi\|_{L^2(\mu)}^2$$

and from this we deduce that

$$\left\| t^{\frac{\alpha b}{2a}} \phi\left(t^{\frac{b}{a}} \cdot\right) \right\|_{L^2(\mu)}^2 = \|\phi\|_{L^2(\mu)}^2 = \|f\|_{\mathcal{H}}^2$$

and in addition

$$W_n \left(t, t^{\frac{b\alpha}{2a}} \phi\left(t^{\frac{b}{a}} \cdot\right) \right) = t^{n\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} W_n(1, \phi)$$

and the result follows after taking the sup over all $f \in \mathcal{H}$ with $\|f\|_{\mathcal{H}} = 1$. □

Lemma 6.7.

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} U_n(1) \right] \geq \log \left[\frac{\rho^{1/2}}{\left(\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)\right)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right]$$

Proof: Let τ be an exponential random variable with mean 1. By lemma 6.7,

$$\mathbb{E}(U_n(\tau)) = \mathbb{E} \left(\tau^{n\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} U_n(1) \right) = \Gamma \left(n\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right) U_n(1)$$

and by Theorem 6.6 we obtain

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[\Gamma \left(n\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right) U_n(1) \right] \geq \log(\rho^{1/2}). \quad (6.7)$$

Recall that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{\Gamma(an+1)}{(n!)^a} \right) = a \log(a)$$

and with this we have that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{\Gamma \left(n\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right)}{(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}} \right] = \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log \left(\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right)$$

and the result is proven after multiplying and dividing $(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}$ on the inside of the log in [6.7](#). □

Lemma 6.8. *For any $k, \theta > 0$, there exists a constant $c_1 = c_1(\alpha, \mathcal{M}, k, \theta) > 0$ such that, by setting $n_t = \lceil c_1 t \rceil$, it holds that*

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log(k^{n_t} \theta^{n_t/2} U_{n_t}(t)) \geq \left(k\sqrt{\theta} \right)^{\frac{2a}{2ab-\alpha b+2ar}} \left(\rho^{1/2} \right)^{\frac{2a}{2ab-\alpha b+2ar}} \quad (6.8)$$

Proof: Let $\epsilon > 0$. Lemma 6.8 guarantees the existence of an $N_\epsilon > 0$ so for all $n > N_\epsilon$

$$(n!)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)} U_n(1) \geq \exp(n(\log(R) - \epsilon)) = R^n e^{-n\epsilon}$$

where

$$R = \frac{\rho^{1/2}}{\left(\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)\right)^{\frac{b}{a}(a-\frac{\alpha}{2}+\frac{a}{b}r)}}.$$

Now fix a $c > 0$ and let $n_t := [ct]$. Notice that $n_t \geq N_\epsilon$ for any $t > t_\epsilon := (N_\epsilon + 1)/c$.

Let $t > t_\epsilon$.

$$k^{n_t} \theta^{n_t/2} U_{n_t}(t) = k^{n_t} \theta^{n_t/2} t^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} U_{n_t}(1) \geq \frac{k^{n_t} \theta^{n_t/2} t^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} R^{n_t} e^{-n_t \epsilon}}{(n_t!)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}}$$

Notice that $[ct]/t \rightarrow c$ as $t \rightarrow \infty$ which means that $n_t/t \rightarrow c$ as $t \rightarrow \infty$. With this we can say

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log \left(k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) = c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left(k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) =: I(n_t)$$

Now,

$$\begin{aligned} I(n_t) &\geq c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left((kR\sqrt{\theta})^{n_t} \frac{t^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} n_t^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}}{(n_t!)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) + c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left[\left(\frac{t}{n_t} \right)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} \frac{n_t^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}}{(n_t!)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}} \right] - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left(\frac{n_t^{n_t}}{(n_t)!} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) - c\epsilon \end{aligned}$$

and letting ϵ tend to 0 we see that

$$I(n_t) \geq c \left[\log(k\sqrt{\theta}R) - \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right] =: h(c).$$

We can easily find the maximal c by solving $h'(c) = 0$ and doing so we get $c = (k\sqrt{\theta}R)^{\frac{a}{b(a - \frac{\alpha}{2} + \frac{a}{b}r)}}$. After plugging in this value for c and replacing R we arrive at the following inequality

$$I(n_t) \geq (k\sqrt{\theta})^{\frac{a}{b(a - \frac{\alpha}{2} + \frac{a}{b}r)}} (\rho_{new}^{1/2})^{\frac{a}{b(a - \frac{\alpha}{2} + \frac{a}{b}r)}}$$

and simplifying exponents gives us 6.8. □

We now can prove Theorem 6.1.

Proof of Theorem 6.1: By proposition 6.4, for and $p, q > 0$ with $p^{-1} + q^{-1} = 1$ we have that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \right|.$$

We now take the supremum over all $f \in \mathcal{H}_+$ with $\|f\|_{\mathcal{H}} = k > 0$. Recall that $W_n(t, \phi) = k^n W_n(t, \phi/k)$ for $\phi \in L^2(\mu)$ and non-negativity of $W_n(t, \cdot)$ on $\mathcal{H}_+$. Let $c > 0$.

$$\begin{aligned} \|u(t, 0)\|_p &\geq \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}} = k} \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \\ &= \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}} = 1} \sum_{n \geq 0} k^n \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \\ &\geq \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}} = 1} k^{n_t} \theta^{n_t/2} U_{n_t}(t_p^\beta, f) \end{aligned}$$

where $n_t = \lceil ct_p^\beta \rceil$. Now by choosing c as in Lemma 6.8 we get that

$$\begin{aligned} \liminf_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p &\geq \liminf_{n \rightarrow \infty} \left(\frac{-\frac{1}{2} t_p^\beta k^2}{t_p^\beta} + \frac{1}{t_p^\beta} \log \left[\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} k^{n_t} \theta^{n_t/2} U_{n_t}(t_p^\beta, f) \right] \right) \\ &\quad - \frac{1}{2} k^2 + (k\sqrt{\theta})^{\frac{2a}{2ab-\alpha b+2ar}} (\rho^{1/2})^{\frac{2a}{2ab-\alpha b+2ar}} =: h(k) \end{aligned}$$

By maximizing h we see that h is maximized at the point $k^* = \left(\frac{2ab-\alpha b+2ar}{2aB} \right)^{\frac{2ab-\alpha b+2ar}{2a-2[2ab-\alpha b+2ar]}}$ where $B = (\sqrt{\theta} \rho^{1/2})^{\frac{2a}{2ab-\alpha b+2ar}}$. Inserting k^* into h gives us that

$$h(k^*) = B^\beta \left(\frac{2a}{2ab-\alpha b+2ar} \right)^\beta \left(\frac{2ab-\alpha b+2ar-a}{2a} \right)$$

and plugging in the value for B yields 6.1. □

References

- [1] Balan, R. M., Chen, L. and Chen, X. (2021). Exact asymptotics of the stochastic wave equation with time-independent noise. *Ann. Inst. Henri Poincaré: Probab. & Stat.*, to appear.
- [2] Balan, R. M. and Song, J. (2017). Hyperbolic Anderson Model with space-time homogeneous Gaussian noise. *ALEA, Latin Amer. J. Probab. Math. Stat.* **14**, 799–849.
- [3] Chen, L., Hu, Y. and Nualart D. (2019). Nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$. *Science Direct, Stochastic Processes and Their Applications.* **129**, 5073-5112
- [4] Chen X. *Parabolic Anderson model with rough or critical Gaussian noise.* (2019) Ann. Inst. Henri Poincaré: Prob. Stat. **55**, 941–976.
- [5] Bass, R., Chen, X. and Rosen, M. (2009). Large deviations for Riesz potentials of additive processes. Ann. Inst. Henri Poincaré: Probab. & Stat. **45**, 626-666
- [6] Chen, L., Hu, Y., Hu, G. and Huang, J. *Stochastic time-fractional diffusion equations on $\mathbb{R}^d$* Stochastics **89** (1) (2007) 171-206
- [7] Dalang, Robert, Davar Khoshnevisan, Carl Mueller, David Nualart, and Yimin Xiao *A minicourse on stochastic partial differential equations.* Held at the University of Utah, Salt Lake City, UT, May 819, 2006. Edited by Khoshnevisan and Firas Rassoul-Agha. Lecture Notes in Mathematics, 1962. Springer-Verlag, Berlin, 2009. xii+216 pp.
- [8] Hu, Y. Hu, Y. (2001) Heat equation with fractional white noise potential. *Appl. Math. Optim.* **43**, 221-243.
- [9] Nualart D. and Nualart E. (2018) Introduction to Malliavin Calculus. Cambridge University Press.